

DEPARTEMENT STATISTIEK EN AKTUARIËLE WETENSKAP
DEPARTEMENT OF STATISTICS AND ACTUARIAL SCIENCE

STATISTIEK EN DATAWETENSKAP 188
STATISTICS AND DATA SCIENCE 188

TUTORIAAL 6 / TUTORIAL 6

1. Gegee die standaard normaal verdeling [dit is, die Normaal($\mu = 0$; $\sigma = 1$) verdeling], wat is die waarskynlikheid dat:
Given a standardised normal distribution [that is, a Normal($\mu = 0$; $\sigma = 1$) distribution] what is the probability that:
 - a. Z minder is as 1.57? / Z is less than 1.57?
 - b. Z minder is as 1.84? / Z is less than 1.84?
 - c. Z groter is as 1.84? / Z is greater than 1.84?
 - d. Z tussen 1.57 en 1.84 is? / Z is between 1.57 and 1.84?
 - e. Z minder is as 1.57 of meer is as 1.84? / Z is less than 1.57 or greater than 1.84?
2. Gegee die standaard normaal verdeling, wat is die waarskynlikheid dat: /
Given a standardised normal distribution, what is the probability that:
 - a. $P(Z > 1.08)$?
 - b. $P(Z < -0.21)$?
 - c. $P(-1.96 < Z < -0.21)$?
 - d. Wat is die waarde van Z as 15.87 persent van alle moontlike Z -waardes groter is? /
What is the value of Z if only 15.87 percent of all possible Z values are larger?
3. Gegee die normaal verdeling met $\mu = 50$ en $\sigma = 4$, bereken die volgende waarskynlikhede:
Given a normal distribution with $\mu = 50$ and $\sigma = 4$, estimate the following probabilities:
 - a. $P(X > 43)$?
 - b. $P(X < 42)$?
 - c. Vyf persent van die waardes is kleiner as watter X -waarde? /
Five percent of the values are less than what X value?
 - d. Tussen watter twee X -waardes (simmetries verdeel rondom die gemiddelde) is 60 persent van die waardes? /
Between what two X values (symmetrically distributed around the mean) are 60 percent of the values?

4. 'n Stel finale kursus punte in 'n inleidende statistiek kursus is normaal verdeel, met 'n gemiddelde van 73 (uit 100) en 'n standaardafwyking van 8. /
A set of final course grades in an introductory statistics course is normally distributed, with a mean of 73 (out of 100) and a standard deviation of 8.
- Wat is die waarskynlikheid dat 'n student minder as 91 behaal het in die finale punt? /
What is the probability that a student earned below 91 as the final grade?
 - Wat is die waarskynlikheid dat 'n student 'n finale punt tussen 65 en 89 behaal het? /
What is the probability that a student earned a final grade between 65 and 89?
 - Die waarskynlikheid is 5 persent dat 'n student wat die kursus volg, hoër verdien as watter punt? /
The probability is 5 percent that a student that takes the course earns higher than what grade?
 - Die universiteit gebruik 'n kurwe-graderingstelsel, sodanig dat tien persent van die studente wat vir 'n kursus geregistreer 'n onderskeiding kry, ongeag die punt. Is jy beter af as jy 'n finale punt van 81 kry vir hierdie kursus, of as jy 'n finale punt van 68 kry vir 'n aparte kursus met 'n gemiddelde van 62 en 'n standaardafwyking van 3? Verduidelik jou antwoord? /
The university operates a curve grading system such that the top ten percent of the students registered for a course gets a distinction, regardless of the score. Are you better off earning a final grade of 81 for this course or earning a final grade of 68 for a separate course that has a mean of 62 and a standard deviation of 3? Explain your answer?
5. Baie vervaardigingsprobleme behels die aanpassing van masjienonderdele, soos skagte wat in 'n klepgat pas. 'n Spesifieke ontwerp vereis 'n as met 'n deursnee van 22.000 mm, maar skagte met 'n diameter tussen 21.990 en 22.010 mm is aanvaarbaar. Gestel die vervaardigingsproses lewer skagte met deursnee wat normaal versprei is, met 'n gemiddelde van 22.002 mm en 'n standaardafwyking van 0.005 mm. Vir die proses, wat is: /
Many manufacturing problems involve the matching of machine parts, such as shafts that fit into a valve hole. A particular design requires a shaft with a diameter of 22.000mm, but shafts with diameters between 21.990mm and 22.010mm are acceptable. Suppose that the manufacturing process yields shafts with diameter normally distributed, with a mean of 22.002mm and a standard deviation of 0.005mm. For this process, what is:
- die waarskynlikheid dat 'n skag aanvaarbaar is? /
the probability that a shaft is acceptable?
 - die deursnee wat slegs twee persent van die skagte oorskry? /
the diameter that will be exceeded by only two percent of the shafts?
 - Wat sou jy antwoorde in (a) tot (c) wees as die standaardafwyking van die asdiameters 0.004mm was? /
What would be your answers in (a) through (c) if the standard deviation of the shaft diameters were 0.004mm?

6. Gee die ses Z-waardes vir 'n steekproef van ses eweredig verdeelde standaard normaal waarnemings. /
List the six Z values for a sample of six evenly distributed standard normal observations.
7. Toon aan dat die kleinste en grootste waardes onderskeidelik -1.96 en +1.96 is vir 'n steekproef van 39 waarnemings wat eweredig versprei is en wat uit 'n standaard normaal verdeling geneem is. /
Show that for a sample of evenly spread 39 observations that are drawn from a standard normal distribution, the smallest and largest values are -1.96 and +1.96, respectively.
8. Gestel jy kies een waarde uit 'n uniforme verdeling met $a = 0$ en $b = 10$. Wat is die waarskynlikheid dat die waarde: /
Suppose that you select one value from a uniform distribution with $a = 0$ and $b = 10$. What is the probability that the value will be:
 - a. tussen 5 en 7 is? / *between 5 and 7?*
 - b. tussen 2 en 3 is? / *between 2 and 3?*
 - c. Wat is die gemiddelde? / *What is the mean?*
 - d. Wat is die standaardafwyking? / *What is the standard deviation?*
9. Hoe lank neem dit om 'n HD-fliek van twee uur af te laai vanaf 'n media-werf? Volgens Fastmetric moet die aflaai van so 'n fliek met 'n internetverbinding van 100 Mbps tussen 3.5 en 5.5 minute duur. Gestel jy oorweeg dit om een van sulke flieks onder soortgelyke internettoestande af te laai. Wat is die waarskynlikheid dat die aflaai tyd: /
How long does it take to download a two-hour HD movie from a streaming media site? According to Fastmetric, downloading such a movie using a 100Mbps internet connection should take between 3.5 and 5.5 minutes. Consider downloading one of such movies under similar internet condition. What is the probability that the download time will be:
 - a. minder as 4.5 minute sal wees? / *less than 4.5 minutes?*
 - b. meer as 4.0 minute sal wees? / *more than 4.0 minutes?*
 - c. tussen 4.0 en 4.5 minute sal wees? / *between 4.0 and 4.5 minutes?*
 - d. Wat is die gemiddelde en standaardafwyking van die aflaai tye? /
What are the mean and standard deviation of the download times?
10. Gegee 'n eksponensiaal verdeling met $\lambda = 10$, wat is die waarskynlikheid dat die arriverings tyd: /
Given an exponential distribution with $\lambda = 10$, what is the probability that the arrival time is:
 - a. minder as 0.1 is? / *less than 0.1?*
 - b. groter as 0.1 is? / *greater than 0.1?*
 - c. tussen 0.1 en 0.2 is? / *between 0.1 and 0.2?*
 - d. minder as 0.1 of groter as 0.2 is? / *less than 0.1 or greater than 0.2?*

11. Gegee dat tyd 'n eksponensiaal verdeling het met $\lambda = 5$, wat is die waarskynlikheid dat die arrivering tyd: /
Given an exponential distribution with $\lambda = 5$, what is the probability that the arrival time is:
- minder as 0.3 is? / *less than 0.3?*
 - groter as 0.3 is? / *greater than 0.3?*
 - groter as 0.3 en minder as 0.5 is? / *greater than 0.3 and less than 0.5?*
 - minder as 0.3 of groter as 0.5 is? / *less than 0.3 or greater than 0.5?*
12. Motors arriveer by 'n tolplein by die ingang van 'n brug teen 'n snelheid van 50 per minuut gedurende die 17h00 en 18h00 uur. As 'n motor pas aangekom het: /
Cars arrive at a toll plaza located at the entrance to a bridge at a rate of 50 per minute during the 17h00 and 18h00 hour. If a car just arrived:
- wat is die waarskynlikheid dat die volgende motor binne 3 sekondes (0.05 minute) sal arriveer? /
what is the probability that the next car will arrive within 3 seconds (0.05 minutes)?
 - wat is die waarskynlikheid dat die volgende motor binne die volgende sekonde sal arriveer? /
what is the probability that the next car will arrive within the next second?
 - Hoe sal die antwoorde in (a) en (b) verander as die aankoms van motors 60 per minuut is? /
How will the answers in (a) and (b) change if the rate of arrival of cars is 60 per minute?
 - Hoe sal die antwoorde in (a) en (b) verander as die aankoms van motors 30 per minuut is? /
How will the answers in (a) and (b) change if the rate of arrival of cars is 30 per minute?
13. Oproepe arriveer teen 'n koers van 12 per uur by 'n oproepsentrum. Wat is die waarskynlikheid dat die volgende oproep in: /
Calls arrive at a call centre at the rate of 12 per hour. What is the probability that the next call arrives in:
- minder as 1 minuut sal arriveer? / *less than 1 minute?*
 - minders as 3 minute sal arriveer? / *less than 3 minutes?*
 - meer as 6 minute sal arriveer? / *more than 6 minutes?*
14. Vir $n = 100$ en $\pi = 0.20$, gebruik die normaal verdeling en benader die waarskynlikheid dat: /
For $n = 100$ and $\pi = 0.20$, use the normal distribution to approximate the probability that:
- $X = 25$
 - $X < 25$
 - $X \leq 25$
 - $X > 25$

15. Vir $n = 100$ en $\pi = 0.40$, gebruik die normaal verdeling en benader die waarskynlikheid dat: /
For $n = 100$ and $\pi = 0.40$, use the normal distribution to approximate the probability that:
- $X = 40$
 - $X < 40$
 - $X \leq 40$
 - $X > 40$
16. Beskou 'n eksperiment waar jy 'n regverdige muntstuk 12 keer gooi en die aantal koppe ("heads") tel. Verkry die waarskynlikheid van: /
Consider an experiment in which you toss a fair coin 12 times and count the number of heads. Determine the probability of observing:
- 4 koppe / heads.
 - ten minste 4 koppe / *at least 4 heads.*
 - 5, 6 of / *or 7 koppe / heads.*
 - Gebruik die normaal benadering tot die binomiaal verdeling om die waarskynlikhede in (a), (b) en (c) te benader. /
Use the normal approximation to the binomial distribution to approximate the probabilities in (a), (b) and (c).
17. Vir kort binnelandse vlugte het 'n lugredery drie verskillende keuses op sy peuselhappiespyskaart - pretzels, aartappelskyfies en koekies. Op grond van vorige ervaring, is die lugredery van mening dat elke peuselhappie gelyke kans het om gekies te word. As daar 150 passasiers op 'n spesifieke vlug is, wat is die benaderde waarskynlikheid dat: /
For short domestic flights, an airline has three different choices on its snack menu – pretzels, potato chips and cookies. Based on past experience, the airline feels that each snack is equally likely to be chosen. If there are 150 passengers on a particular flight, what is the approximate probability that:
- presies 60 passasiers sal pretzels vir nagereg kies? /
exactly 60 passengers will choose pretzels for dessert?
 - minstens 60 passasiers pretzels vir nagereg kies? /
at least 60 passengers will choose pretzels for dessert?
 - minder as 60 passasiers pretzels vir nagereg sal kies? /
fewer than 60 passengers will choose pretzels for dessert?
 - As die lugredery 70 van elke soort peuselhappie op die vlug beskikbaar het, wat is die waarskynlikheid dat 'n passasier nie pretzels vir nagereg kan kry nie? / *If the airline has 70 of each type of snack available on the flight, what is the probability that at least one passenger will not be able to get pretzels for dessert?*

Microsoft Excel® Questions

18. Een bewerking van 'n meul is om stukke staal in dele te sny wat later as raamwerk vir die voorste sitplekke in 'n motoraanleg gebruik word. Die staal word met 'n diamantsaag gesny en die dele wat daaruit voortkom, moet binne ± 0.005 duim van die lengte wat deur die motoronderneming gespesifiseer word, wees. Die data kom uit 'n steekproef van 100 staalonderdele en word in die **Steel** - lêer gestoor. Die meting wat gerapporteer is, is die verskil in duim tussen die werklike lengte van die staaldeel, gemeet deur 'n lasermetingtoestel, en die gespesifiseerde lengte van die staaldeel. Bepaal of die data ongeveer normaal verdeel is deur: /

*One operation of a mill is to cut pieces of steel into parts that will later be used as the frame for front seats in an automotive plant. The steel is cut with a diamond saw, and the resulting parts must be within ± 0.005 inch of the length specified by the automobile company. The data come from a sample of 100 steel parts and are stored in the **Steel** file. The measurement reported is the difference, in inches, between the actual length of the steel part, as measured by a laser measurement device, and the specified length of the steel part. Determine whether the data appear to be approximately normally distributed by:*

- die data-eienskappe met teoretiese eienskappe te vergelyk. /
comparing data characteristics to theoretical properties.*
 - 'n normaal waarskynlikheidstipping te konstrueren. /
constructing a normal probability plot.*
 - 'n histogram te konstrueer. /
constructing a normal histogram.*
19. Die volgende **Utility** data stem ooreen met die waterkoste in Rand gedurende 'n somermaand van 2019 vir 'n ewekansige steekproef van 50 woonstelle in die suidelike voorstad van Kaapstad. /

*The following **Utility** data correspond to the water costs, in Rands, during a summer month of 2019 for a random sample of 50 self-contained apartments in the southern suburb of Cape Town.*

96	171	202	178	147	102	153	197	127	82
157	185	90	116	172	111	148	213	130	165
141	149	206	175	123	128	144	168	109	167
95	163	150	154	130	143	187	166	139	149
108	119	183	151	114	135	191	137	129	158

Besluit of die data ongeveer normaal verdeel is deur: /

Decide whether the data appear to be approximately normally distributed by:

- a. die data-eienskappe met teoretiese eienskappe te vergelyk. /
comparing the empirical characteristics to theoretical properties.
- b. 'n normaal waarskynlikheidstipping te konstrueren. /
constructing a normal probability plot.
- c. 'n histogram te konstrueer. /
constructing a normal histogram.

20. Die aandbestuurder van 'n restaurant was baie bekommerd oor die tydsduur wat sommige klante in die ry gewag het om te sit. Sy was ook bekommerd oor die sittyte - dit wil sê die tydsduur tussen die sit van 'n klant en die tyd dat sy of hy die restaurant verlaat. In die verloop van een week is 100 klante (met nie meer as een per partytjie nie) ewekansig gekies en hulle wag- en siture (in minute) is in die **Wait** - lêer aangeteken./

*The evening manager of a restaurant was very concerned about the length of time some customers were waiting in line to be seated. She also had some concern about the seating times – that is, the length of time between when a customer is seated and the time she or he leaves the restaurant. Over the course of one week, 100 customers (with no more than one per party) were randomly selected and their waiting and seating times (in minutes) were recorded in the **Wait** file.*

- a. Konstrueer 'n histogram en 'n normaal waarskynlikheidstipping van die wagtye op. Dink u hierdie wagtye lyk meer soos 'n uniforme, 'n eksponensiële of 'n normale verdeling? /
Construct a histogram and a normal probability plot of the waiting times. Do you think these waiting times more closely resemble a uniform, an exponential, or a normal distribution?
- b. Konstrueer 'n histogram en 'n normaal waarskynlikheidstipping van die sittyte op. Dink u hierdie sittyte lyk meer soos 'n uniforme, 'n eksponensiële of 'n normale verdeling? /
Construct a histogram and a normal probability plot of the seating times. Do you think these seating times more closely resemble a uniform, an exponential, or a normal distribution?

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